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cProject: POS Accessories CLI

I2-GIC 1B

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Academic Year 2025-2026

POS Accessories CLI Report

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Institute of Technology of Cambodia

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2025 – 2026

1. Introduction

POS Accessories CLI is a point-of-sale accessories that entirely works on the command line via user choices based on certain interaction with specific actions through a terminal interface that leads towards the solution of what he intends to do. This has been development and implementation of a C++ and the ncurses library program that represents the user with a menu on where he picks the choices from, and as a result with a transaction receipt of user orders.

2. What is CLI?

A Command Line Interface (CLI) is a text based user interface purposely using a terminal directly that allows users to interact with a program to provide much efficiency and flexible on low end computer to life. Unlike graphic user interfaces (GUI) a CLI processes user input through typed commands and displays based on text.

3. Objective

- The main objective of this project is to develop a Point of Sale (POS) Accessories Command Line Interface (CLI) system that enables users to manage accessories products and perform sales transactions efficiently through a terminal-based environment.
- ❖ Specific objectives of the project are:
 - To develop a menu-driven POS system using C++ and the ncurses library for an interactive terminal user interface.
 - To implement a stack data structure using a linked list to manage shopping cart items and transaction receipts.
 - To store and retrieve product information from CSV files for efficient product management and data persistence.
 - To provide product management functions such as searching, updating, and deleting product records.
 - To integrate QR code generation and online payment processing using qrencode and a Flask backend server.
 - To improve understanding of data structures, file handling, terminal-based interfaces, and client-server communication.
 - To simulate a real-world accessories store transaction system through a lightweight and efficient CLI application.

4. Progression

The development of this POS Accessories CLI followed through 4 phases of progression:

Phase 1: Planning and Design

On this project we focus on specific programs for easier management (C++, ncurses, qrencode) and distributing tasks among team members.

Phase 2: Core

Implementation of the main structured terminal UI using ncurses, Stack linked list this case we use this method to display items on the receipt, and products as .csv extension.

Phase 3: Feature Integration

Integration of QR cod via libqrencode-dev (Third-Party) for payment transaction, that cause Flask backend server to handle the user processed.

Phase 4: Testing and Presentation

The final test by my team members and understanding other work which consider prepared for the live presentation slide.

5. System Architecture

The POS Accessories CLI system provides a local client server architecture with a local terminal both frontend and the backend user will be processed for their payment.

Frontend (C++ / ncurses / qrencode)

- Terminal structured UI (CLI) rendered using ncurses library
- Payment of user's transaction from QR Code or Cash
- Mouse interaction events for navigate inventory
- Color theme for both user and seller experience
- Receipt with live preview on the right side

Backend (Python / Flask)

- Deployed on the Vercel Cloude
- Online payment transaction
- Message after making transaction

Key bind:

q – exit program

P – print QR Code

Space bar – Add item

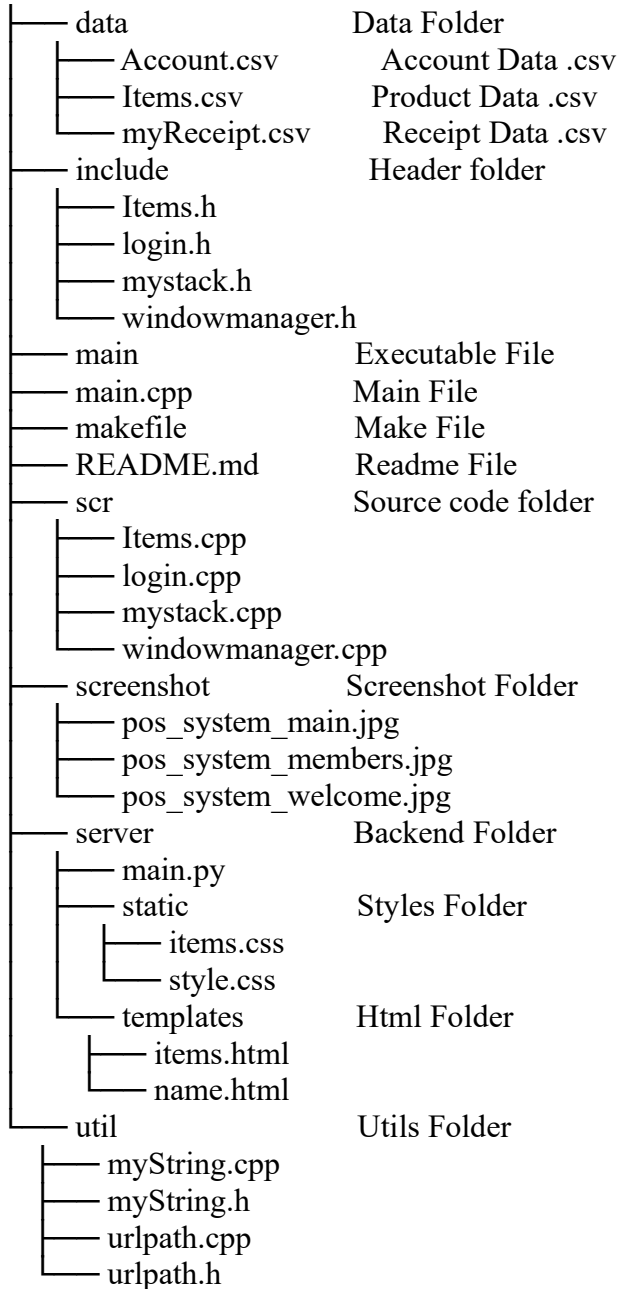
Arrow (up, down, right left) - Navigation

Backspace – Delete item

Enter – save receipt

TREE Diagram

POS



6. Data Management

The system is built using C++. The two main parts of data management are stack structure which uses to handle shopping cart items during transaction and Production (Product information) is stored in csv file which describes all products available in the store.

6.1. Stack Structure

In this system, the stack stores Product objects. The primary operations are:

push: Adds a new item to the top of the stack.

pop: Removes the item from the top of the stack and returns it.

checkExist: Checks whether the item with a given ID already exists in the stack.

searchProduct: Finds the returns an item from the stack by its ID.

updateProduct: Changes the name, size, price, or quantity of an existing item.

deleteProduct: Reduces item quantity by one. Removes the item fully when quantity reaches zero.

Summation: calculates the total price of all items currently in the stack.

Tax: Computes a 10% tax on the total amount.

6.2. Production

The products available in the POS CLI system are accessories. All product data is stored in a file name Items.csv.

Each product in this system contains the following information:

Name: The product name (Watch, Ring, Necklace...)

ID: A unique number assigned to each product.

Price: The selling price in US dollars (\$)

Size: The size of the product example: small (S), Medium (M) or large (L)

7. Advantages and Disadvantages of POS CLI

Advantages of POS CLI:

- Speed and efficiency: POS CLI are very fast and gets the job done quickly. It saves a lot of time for both customers and cashiers
- Better management: It is easy to manage sales and reports through commands, better order
- Required fewer systems
- Low-level interaction between software and hardware
- Reliability and consistency: CLI produce identical results, reducing human error
- Better performance: POS CLI runs very well even on basic and older computers because it required fewer systems

Disadvantages of POS CLI:

- Cannot display pictures: POS CLI is not user friendly it doesn't have pictures. It works only with text which means customers cannot see photos of product they are buying during checkout.
- Hard to learn: POS CLI is not easy for people who aren't used to work with computers. Workers need to memorize many commands and it takes a lot of time for training.

8. Key Benefits

- **Faster checkout times:** providing barcode scanners and integrated payment terminals and receipt printers.
- **Enhance payment Flexibility:** allowing customers to pay in their preferred way such as cash, bank card.
- **Reduced Error:** Automated systems ensure accurate pricing and calculation, reducing errors in billing, change calculation, and inventory, which leads to a smoother, error-free experience.
- **Improved Service & Information:** Customer-facing displays allow customers to view items, prices, and totals in real-time as they are scanned, ensuring transparency and accuracy.
- **Seamless Returns and Exchanges:** Integrated POS systems make returning or exchanging items quick and efficient.
- **Personalized Experience:** Many POS systems track purchase history and customer data, loyalty programs, and personalized offers

9. Installation

The POS Accessories CLI can be installed followed the command line based on specific hardware:

Step 1: Install Dependencies

Arch Linux

```
yay -S ncurses qrencode make git g++  
git clone https://github.com/TinaGrim/POS_accessories_CLI.git  
cd POS_accessories_CLI
```

Debian

```
sudo apt update && sudo apt install libncurses-dev libqrencode-dev make git g++  
git clone https://github.com/TinaGrim/POS_accessories_CLI.git  
cd POS_accessories_CLI
```

Window (Not recommended)

Run this inside msys2 ucrt

```
pacman -S mingw-w64-ucrt-x86_64-gcc make git mingw-w64-ucrt-x86_64-ncurses  
git clone https://github.com/TinaGrim/POS_accessories_CLI.git  
cd POS_accessories_CLI
```

MacOS

```
brew install ncurses qrencode make git g++  
git clone https://github.com/TinaGrim/POS_accessories_CLI.git  
cd POS_accessories_CLI
```

Step 2: Execution

```
make run && make clean
```

10. Conclusion

- The POS Accessories CLI System was successfully developed using C++ and the ncurses library.
- The system allows users to manage accessories products and perform sales transactions through a terminal-based interface.
- A stack data structure implemented using a linked list was used to manage shopping cart items and receipts.
- Product information is stored and retrieved using CSV files, allowing data to be saved even after the program is closed.
- The integration of QR code payment and a Flask backend server provides an additional payment option for users.
- The project helped improve our understanding of data structures, file handling, terminal user interfaces, and client-server communication.
- The system is lightweight, efficient, and can run on computers with limited resources.
- Although the system has some limitations, such as a text-based interface and limited graphical features, it successfully meets the objectives of the project.
- In the future, the system can be improved by adding a database, user authentication, sales reports, and a graphical user interface (GUI).

10.1. References

C++ – <https://www.stroustrup.com/>

Ncurses – <https://invisible-island.net/ncurses/>

Msys2 – <https://www.msys2.org/>

Makefile – <https://dev.to/shrsv/mastering-makefiles-from-beginner-basics-to-pro-level-patterns-and-tricks-p72>

Qrencode – <https://github.com/fukuchi/libqrencode>